

SANDWICHED SINGULARITIES AND PLANE CURVE ARRANGEMENTS

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ABSTRACT. These notes provide a mathematically rigorous, self-contained exposition on the geometry, resolution graphs, and deformation theory of sandwiched surface singularities, specifically focusing on the framework utilized by Plamenevskaya and Starkston (2020) and de Jong–van Straten (1998). We unpack the equivalence between the analytical definition of a rational surface singularity with a reduced fundamental cycle (RFC) and the numerical combinatorial constraints imposed on its dual resolution tree G . By formalizing the algorithmic construction of the expanded plumbing graph G' , we explicitly trace the introduction of transverse curvetas along distinguished (-1) -exceptional curves. Finally, we establish the explicit dictionary between the multiplicity sequences of algebraic point blow-ups on smooth curve germs in $(\mathbb{C}^2, 0)$ and the weight functions $w(C_i)$ or decorations $l(i)$ required to reconstruct the analytic space $X(C, l)$ of the singularity.

CONTENTS

1	Reduced Fundamental Cycle	1
2	Sandwiched Singularities	2
3	Curves and Curvetas	6

1. REDUCED FUNDAMENTAL CYCLE

Definition 1.1. Let $(X, 0)$ be a rational surface singularity, and let $\pi : \tilde{X} \rightarrow X$ be its minimal normal crossings resolution with exceptional curves $E_1, \dots, E_n$. The *fundamental cycle* is the unique minimal positive divisor $Z = \sum_{i=1}^n c_i E_i$ (where $c_i \in \mathbb{Z}_{>0}$) supported on the exceptional set that satisfies the anti-nef condition:

$$Z \cdot E_i \leq 0 \quad \text{for all } i = 1, \dots, n. \quad (1)$$

The singularity $(X, 0)$ is said to have a *reduced fundamental cycle* (RFC) if every coefficient in the fundamental cycle is equal to one, meaning:

$$Z = \sum_{j=1}^n E_j. \quad (2)$$

Let $(X, 0)$ be a rational surface singularity with a reduced fundamental cycle $Z = \sum_{j=1}^n E_j$. We evaluate the intersection product $Z \cdot E_i$ for an arbitrary vertex $v_i \in G$. Expanding this product

yields:

$$Z \cdot E_i = \left(E_i + \sum_{j \neq i} E_j \right) \cdot E_i = E_i^2 + \sum_{j \neq i} (E_j \cdot E_i). \quad (3)$$

Since the resolution is normal crossings, the sum $\sum_{j \neq i} (E_j \cdot E_i)$ counts exactly the number of distinct exceptional components meeting E_i , which is the graph valency $a(v_i)$. Substituting this and the self-intersection $E_i^2 = v_i \cdot v_i$, we obtain:

$$Z \cdot E_i = v_i \cdot v_i + a(v_i). \quad (4)$$

The anti-nef property dictates that $Z \cdot E_i \leq 0$, which implies $v_i \cdot v_i + a(v_i) \leq 0$, or equivalently $v_i \cdot v_i \leq -a(v_i)$. For any connected graph with $n > 1$, every vertex must possess a valency of at least $a(v_i) \geq 1$, which rigorously proves that $v_i \cdot v_i \leq -1$. For a terminal leaf vertex $v_{\text{leaf}} \in G$ where the valency is exactly $a(v_{\text{leaf}}) = 1$, if the curve has a minimal self-intersection of -2 , the product evaluates to exactly:

$$Z \cdot E_{\text{leaf}} = (-2) + 1 = -1. \quad (5)$$

This demonstrates how the fundamental cycle directly hits the upper boundary value of -1 at the terminal ends of the tree.

2. SANDWICHED SINGULARITIES

Definition 2.1. A normal surface singularity $(X, 0)$ is a *sandwiched singularity* if it admits a birational map $\pi : (X, 0) \rightarrow (\mathbb{C}^2, 0)$.

Definition 2.2. Let X be a surface and $p \in X$. Then there exist a surface $\tilde{X}$ and a morphism $\pi : \tilde{X} \rightarrow X$, which are unique up to isomorphism, such that:

1. the restriction of π to $\pi^{-1}(X - \{p\})$ is an isomorphism onto $X - \{p\}$;
2. $\pi^{-1}(p) = E$ is isomorphic to $\mathbb{P}^1$.

The blow-up of $\mathbb{C}^2$ at the origin replaces the origin with a projective line $\mathbb{P}^1$ representing all lines (slopes) through the origin. This space is covered by two distinct coordinate charts mapping down to the original (x,y) plane.

To rigorously trace the algebraic and geometric mechanics of consecutive point blow-ups with full analytic precision, we choose the ordinary cusp germ at the origin in $\mathbb{C}^2$, defined explicitly by the zero locus of the polynomial:

$$f(x, y) = y^2 - x^3 = 0. \quad (6)$$

A point blow-up replaces a single isolated point with a projective line $\mathbb{P}^1 \cong \mathbb{C} \cup \{\infty\}$, parameterizing all possible directional slopes approaching that point. Because an entire projective line cannot be mapped using a single affine coordinate grid, we cover the blown-up surface using two overlapping coordinate charts, tracking how every curve function transforms across both coordinate systems.

For the first blow-up step centered at the original origin $p_0 = (0, 0) \in \mathbb{C}^2$, we define the first chart (U_x) and the second chart (U_y) by their explicit transition morphisms:

$$\pi_1 : U_x \rightarrow \mathbb{C}^2, \quad x = u, \quad y = vu \quad (7)$$

$$\pi_1 : U_y \rightarrow \mathbb{C}^2, \quad x = ts, \quad y = s \quad (8)$$

In the first chart U_x , the parameters u and v represent the horizontal coordinate and the localized directional slope ($v = y/x$), respectively. The exceptional divisor $E_1 = \pi_1^{-1}(0, 0)$ is defined analytically by the vanishing of the collapsing axis coordinate:

$$E_1 \cap U_x = \{(u, v) \in U_x \mid u = 0\}. \quad (9)$$

To calculate the total transform $\pi_1^{-1}(C)$ within U_x , we substitute the equations $x = u$ and $y = vu$ directly into the defining polynomial $f(x, y)$:

$$\pi_1^* f(u, v) = (vu)^2 - (u)^3 = v^2 u^2 - u^3 = u^2(v^2 - u). \quad (10)$$

Setting the total pullback polynomial equal to zero gives the equation for the total transform:

$$u^2(v^2 - u) = 0. \quad (11)$$

This expression factors cleanly into two distinct components. The term $u^2 = 0$ represents the exceptional divisor E_1 carrying an algebraic multiplicity of 2, which perfectly matches the original multiplicity of the cusp at the origin. The remaining factor yields the equation for the proper (strict) transform $\tilde{C}$:

$$\tilde{C} \cap U_x = \{(u, v) \in U_x \mid v^2 - u = 0\}. \quad (12)$$

To ensure no geometric features are missed, we perform the parallel calculation in the alternate chart U_y , where s tracks the vertical axis and t tracks the inverse slope ($t = x/y$). The exceptional divisor E_1 in this domain is given by $s = 0$. Substituting $x = ts$ and $y = s$ into $f(x, y)$ reveals the total transform in this chart:

$$\pi_1^* f(s, t) = (s)^2 - (ts)^3 = s^2 - t^3 s^3 = s^2(1 - t^3 s) = 0. \quad (13)$$

The algebraic factoring yields the exceptional divisor $s^2 = 0$ with multiplicity 2, and the alternate chart proper transform expression $1 - t^3 s = 0$. To analyze the intersection of the proper transform with the exceptional divisor in U_y , we evaluate the simultaneous system:

$$\begin{cases} s = 0 \\ 1 - t^3 s = 0 \end{cases} \implies 1 - t^3(0) = 0 \implies 1 = 0. \quad (14)$$

Because this system yields an empty solution set, the proper transform does not cross the exceptional divisor anywhere within the U_y chart region. Thus, the intersection behavior is fully and cleanly captured inside the U_x chart.

Returning to U_x , we locate the exact intersection point between the proper transform $\tilde{C}$ ($v^2 - u = 0$) and the exceptional divisor E_1 ($u = 0$) by solving their simultaneous equations:

$$\begin{cases} u = 0 \\ v^2 - u = 0 \end{cases} \implies v^2 - 0 = 0 \implies v = 0. \quad (15)$$

This isolates a single, unique intersection point at the new local origin $p_1 = (u, v) = (0, 0)$. At this coordinate, the proper transform equation is a parabola $u = v^2$, meaning its tangent vector aligns perfectly with the tangent vector of the line $u = 0$. Because they are mutually tangent rather than intersecting transversely, this setup violates the conditions for a normal crossings resolution. We must execute a second blow-up step centered precisely at this point $p_1 = (0, 0)$.

We establish a new pair of coordinate charts for the second blow-up layer, setting up local tracking variables. If we attempt to use a chart based on the horizontal axis by setting $u = u_1$

and $v = u_1 v_1$, we substitute these expressions directly into our current proper transform equation ($v^2 - u = 0$):

$$(u_1 v_1)^2 - (u_1) = u_1^2 v_1^2 - u_1 = u_1(u_1 v_1^2 - 1) = 0. \quad (16)$$

Factoring out the new exceptional divisor E_2 ($u_1 = 0$) leaves the proper transform equation $u_1 v_1^2 - 1 = 0$. If we check its intersection with E_2 by setting $u_1 = 0$, the equation drops to $-1 = 0$, which contains no mathematical solutions. This algebraic failure proves that the proper transform behaves vertically at this stage and completely exits this specific coordinate chart.

To track the curve, we must use the alternate second-layer chart (U_v), defined by the parameter substitutions:

$$v = v_1, \quad u = u_1 v_1. \quad (17)$$

The new exceptional divisor created by this second blow-up step is defined analytically by:

$$E_2 \cap U_v = \{(u_1, v_1) \in U_v \mid v_1 = 0\}. \quad (18)$$

We now calculate how both existing curve components transform under these new coordinates by substituting them into their respective equations from the previous layer:

1. **Proper Transform $\tilde{C}$:** Substituting $v = v_1$ and $u = u_1 v_1$ into $v^2 - u = 0$ gives:

$$(v_1)^2 - (u_1 v_1) = v_1^2 - u_1 v_1 = v_1(v_1 - u_1) = 0. \quad (19)$$

Factoring out the exceptional line E_2 ($v_1 = 0$) yields the newly updated proper transform equation:

$$\tilde{C} : v_1 - u_1 = 0 \implies v_1 = u_1. \quad (20)$$

2. **First Exceptional Divisor E_1 :** The original divisor equation was $u = 0$. Substituting $u = u_1 v_1$ directly into this expression gives the total transform of the first divisor:

$$u_1 v_1 = 0. \quad (21)$$

Since $v_1 = 0$ represents the new exceptional divisor E_2 , the remaining distinct factor defines the proper transform of our first exceptional divisor:

$$E_1 : u_1 = 0. \quad (22)$$

Let us evaluate the geometric state of our system inside this chart by checking the mutual intersection point of all three active components: E_2 ($v_1 = 0$), E_1 ($u_1 = 0$), and $\tilde{C}$ ($v_1 - u_1 = 0$). Solving their equations simultaneously yields:

$$\begin{cases} v_1 = 0 \\ u_1 = 0 \\ v_1 - u_1 = 0 \end{cases} \implies 0 - 0 = 0. \quad (23)$$

This system vanishes perfectly at the point $p_2 = (u_1, v_1) = (0, 0)$. While each component is now a completely smooth line and they intersect each other at non-zero angles (transversely), all three lines pass through the exact same point p_2 . This arrangement forms an algebraic triple point, which violates the strict requirements of a normal crossings configuration. To resolve this, we must execute a third point blow-up centered directly at $p_2 = (0, 0)$.

We define the coordinate transformations for the third blow-up layer. Choosing the chart where the horizontal component acts as the base parameter, we apply the substitutions:

$$u_1 = u_2, \quad v_1 = u_2 v_2. \quad (24)$$

The brand new exceptional divisor generated by this third step is defined analytically by:

$$E_3 = \{(u_2, v_2) \mid u_2 = 0\}. \quad (25)$$

We track the analytic behavior of all three intersecting components by substituting these third-layer parameters into each of their previous equations:

1. **Proper Transform $\tilde{C}$:** The equation from the second layer was $v_1 - u_1 = 0$. Substituting our new coordinates gives:

$$(u_2 v_2) - (u_2) = u_2 v_2 - u_2 = u_2(v_2 - 1) = 0. \quad (26)$$

Factoring out the new exceptional line E_3 ($u_2 = 0$) isolates the fully resolved proper transform of our curve:

$$\tilde{C} : v_2 - 1 = 0 \implies v_2 = 1. \quad (27)$$

2. **Second Exceptional Divisor E_2 :** The equation from the second layer was $v_1 = 0$. Substituting $v_1 = u_2 v_2$ gives:

$$u_2 v_2 = 0. \quad (28)$$

Factoring out E_3 ($u_2 = 0$) yields the proper transform of the second divisor:

$$E_2 : v_2 = 0. \quad (29)$$

3. **First Exceptional Divisor E_1 :** The equation from the second layer was $u_1 = 0$. Substituting $u_1 = u_2$ gives:

$$E_1 : u_2 = 0. \quad (30)$$

This shows that inside this specific coordinate chart, the proper transform of E_1 matches the locus of the new exceptional divisor E_3 .

Let us evaluate the final geometric spacing of our components along the vertical v_2 -axis inside this third-layer space:

- The proper transform of the second exceptional divisor E_2 sits along the line $v_2 = 0$.
- The proper transform of our original curve $\tilde{C}$ sits along the parallel line $v_2 = 1$.

Because $0 \neq 1$, these two loci are separated. They no longer intersect at a single point. Every exceptional curve and proper transform branch now intersects every other component transversely, and no more than two components meet at any single coordinate point. This completes the normal crossings embedded resolution of the cusp curve.

The relationship between a rational singularity with a reduced fundamental cycle (RFC) and its characterization as a sandwiched singularity relies on a reverse combinatorial procedure. We do not perform blow-ups directly on the fundamental cycle itself. Instead, we use its numerical values to build a larger graph, denoted G' .

The expanded graph G' represents a smooth surface that can be sequentially contracted (blown down) to a single smooth point. Topologically, a plumbing configuration that contracts to a smooth point always has a 3-sphere (S^3) boundary. The reduced fundamental cycle condition provides an exact recipe for this construction:

- We start with the original resolution graph G of the singularity.
- We use the RFC inequality to determine the exact number of extra lines needed at each vertex: $k_v = -v \cdot v - a(v)$.
- We attach exactly k_v new terminal leaves, each with a self-intersection weight of -1 , to their corresponding vertices.
- This completes the construction of G' . Contracting these newly added -1 lines one by one simplifies the remaining curves until the entire graph collapses down to a smooth point.

This guaranteed combinatorial collapse proves that every rational singularity with a reduced fundamental cycle is automatically sandwiched.

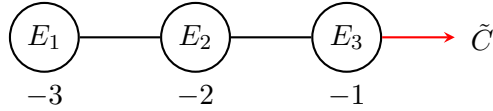


Figure 1: Dual resolution graph for the cusp $y^2 - x^3 = 0$ after three consecutive point blow-ups. The numerical weights represent the self-intersection numbers of each exceptional divisor.

3. CURVES AND CURVETTAS

Definition 3.1. Let $\rho : \tilde{Z} \rightarrow (\mathbb{C}^2, 0)$ be a sequence of point blow-ups with exceptional set $F = \rho^{-1}(0)$. Let T denote the set of distinguished (-1) -curves in $\tilde{Z}$. For each $i \in T$, a *curvetta* $\tilde{C}_i$ is a smooth complex disk chosen transverse to the exceptional curve E_i at a generic point. The union of these disks is denoted $\tilde{C} = \bigcup_{i \in T} \tilde{C}_i$.

Definition 3.2. The *plane curve arrangement* $C \subset (\mathbb{C}^2, 0)$ associated with a sandwiched singularity is the direct image (or pushforward) of the curvetas under the contraction map:

$$C = \rho(\tilde{C}) = \bigcup_{i \in T} C_i, \quad \text{where } C_i = \rho(\tilde{C}_i). \quad (31)$$

The map ρ serves as an embedded resolution of the curve singularity C .

Definition 3.3. Following de Jong and van Straten, a *decorated curve* (or decorated germ) is a pair (C, l) consisting of a plane curve singularity $C = \bigcup_{i \in T} C_i \subset (\mathbb{C}^2, 0)$ and a weighting function $l : T \rightarrow \mathbb{Z}$ that assigns an integer weight to each irreducible branch C_i . The assigned weight tracks the total number of blow-down operations affecting that specific branch and must satisfy the lower bound:

$$l(i) \geq m(i), \quad (32)$$

where $m(i)$ is the sum of multiplicities of branch i in the multiplicity sequence of the minimal embedded resolution of C .

The text establishes a structural restriction to *standard decorated germs* for rational surface singularities with a reduced fundamental cycle (RFC), which guarantees a simplified topology for the expanded graph G' (these are worth noting):

- **Minimal Resolution Conditions:** For an RFC singularity, the minimal normal crossings resolution graph G contains absolutely no (-1) -vertices. By the inequality $a(v) \leq -v \cdot v$, any

vertex with a self-intersection of -1 must have a valency of $a(v) \leq 1$. Such a vertex would be a terminal leaf that could be immediately blown down, violating the definition of a minimal resolution graph.

- **The Leaf Property of Extended Vertices:** Because the minimal resolution graph G is completely devoid of (-1) -vertices, every single (-1) -vertex introduced to construct the expanded graph G' must originate strictly from the extension process.
- **Standard Decorated Germs:** Consequently, each introduced (-1) -vertex acts as a terminal leaf in G' , attached via a single edge to a unique host vertex within G . This prevents complex nested chain extensions of (-1) curves. The transverse curvetta slices are pinned exclusively to these outer leaves, ensuring the corresponding plane curve components C_i are smooth at the origin.

Proof of the (-1) -Vertex Leaf Property

Let G be the dual resolution tree of a rational surface singularity with a reduced fundamental cycle (RFC). We prove that G contains no (-1) -vertices, and that any (-1) -vertex in the expanded graph G' must be a terminal leaf.

Proof. By Definition 2.1, every vertex $v \in G$ must satisfy the reduced fundamental cycle inequality:

$$a(v) \leq -v \cdot v, \quad (33)$$

where $a(v)$ represents the graph valency of v , and $v \cdot v$ denotes its self-intersection number.

Suppose there exists a vertex v such that $v \cdot v = -1$. Substituting this value into the RFC inequality yields:

$$a(v) \leq -(-1) \implies a(v) \leq 1. \quad (34)$$

For any non-trivial connected resolution graph, every vertex must possess a valency of at least 1. Therefore, we are forced to have $a(v) = 1$, which topologically defines v as a terminal leaf.

By Castelnuovo's Criterion, a smooth rational curve with self-intersection -1 and valency 1 can be blown down to a smooth point without violating the normal crossings structure of the remaining exceptional set. By definition, a *minimal* normal crossings resolution cannot contain any contractible exceptional curves. Thus, no such vertex can exist in the minimal graph G .

Consequently, when constructing the expanded graph G' , any vertex with weight -1 must be an introduced extension. Because it carries a weight of -1 , the condition $a(v) \leq 1$ restricts it to being a terminal leaf connected to exactly one host vertex in G . $\square$

To further formalize this structural constraint for our calculations, this leaf property rigorously rules out two complex graph behaviors during the extension from G to G' :

- **No (-1) Chains:** We cannot chain multiple (-1) -vertices together in a line (such as $v_A(-1) - v_B(-1)$). If two (-1) -vertices were connected to each other, vertex v_A would require at least one edge to connect to the main graph G and a second edge to connect to v_B , forcing its valency to be $a(v_A) \geq 2$. This violates the absolute upper bound $a(v) \leq 1$ required by the RFC inequality.
- **No Internal (-1) Nodes:** A (-1) -vertex can never act as a bridging node between two separate original exceptional curves of G . Doing so would require a minimum valency of 2, which is algebraically impossible under the reduced cycle criterion.

This completes the mathematical proof that the extension vertices are strictly isolated, single-edge terminal leaves hanging off the minimal resolution graph G . This exact combinatorial behavior is what defines the configuration as a *standard decorated germ*.

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